SHIVA GOUD

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with hands-on experience developing ESP32-based automotive diagnostic hardware and firmware, CAN communication, and embedded applications using ESP-IDF. Skilled in C/C++, STM32, ARM Cortex-M, bare-metal programming, FreeRTOS, electronics, and PCB design. Builds practical firmware and hardware projects spanning automotive connectivity, embedded cryptography, wireless communication, and real-time systems.

PROFESSIONAL EXPERIENCE

Embedded Systems Developer | Enablero | Mar 2026 – Present | GitHub

- Contributed to the development of wireless automotive diagnostic dongles for Maruti Suzuki, enabling wireless connectivity between a Vehicle Communication Interface (VCI) and a vehicle.
- Worked with ESP32-based hardware on both sides of the link: a VCI-side dongle with a DB26 interface and a vehicle-side dongle with an OBD connector.
- Worked with ESP32, ESP-IDF, CAN communication, and automotive diagnostics as part of the firmware and device-integration workflow.

TECHNICAL SKILLS

Programming: C, Embedded C, C++, ARM Assembly
MCUs & architecture: ESP32, ESP-IDF, STM32, ARM Cortex-M, CMSIS
Embedded development: Bare-metal programming, FreeRTOS, firmware development, interrupt handling
Communication: CAN, UART, SPI, I2C, MQTT, LoRa
Hardware & CAD: Electronics, circuit design, PCB design, KiCad, FreeCAD, SolidCraft
Tools & platforms: Git, GitHub, Linux, GCC, GDB

PROJECTS

Wireless OBD Dongle for Automotive Diagnostics PCB | GitHub

- Developed an ESP32-based wireless dongle concept to bridge a VCI and a vehicle for automotive diagnostics.
- Designed the system around a VCI-side DB26 interface and a vehicle-side OBD connector, with ESP32 hardware providing wireless connectivity.
- Applied embedded firmware development and CAN/automotive diagnostics concepts in a Maruti Suzuki diagnostic use case.

Custom STM32 PCB | GitHub

- Designed a custom STM32-based PCB, covering the hardware design and board-level implementation workflow.
- Used PCB design practices and electronic hardware integration to translate a microcontroller project into a custom board.

ECC/RSA Encryption using PSA Crypto API in ESP-IDF | GitHub

- Implemented device identity and message-security workflows on ESP32 using ESP-IDF and the PSA Crypto API.
- Generated asymmetric keys and certificate signing requests (CSRs), enrolled devices with a CA, and stored CA-signed certificates persistently in NVS.
- Implemented peer-certificate trust validation and an RSA message flow combining public-key encryption/decryption with digital-signature generation and verification.

STM32 LoRa Wireless Messaging System with Display | GitHub

- Built a bidirectional wireless messaging system using STM32 microcontrollers and Ebyte LoRa UART modules.
- Implemented UART-based transmission/reception, interrupt-driven data handling, and buffer management; included an OLED display for message/status feedback.

Computer-Vision Gesture-Controlled Motor and LED System | GitHub

- Developed a real-time hand-gesture interface using MediaPipe, OpenCV, ESP32, and Arduino to control an LED and servo motor.
- Implemented finger-state detection, serial communication to the controller, and OLED status display.

ADDITIONAL PROJECTS

ESP32 FreeRTOS Producer–Consumer Wi-Fi/MQTT | Implemented producer/consumer tasks with queues, mutexes and semaphores, alongside Wi-Fi and MQTT connectivity. GitHub

Google Home Voice-Controlled ESP32 Automation | Integrated ESP32, Sinric Pro and Google Assistant to control a relay-switched light using voice commands. GitHub

EDUCATION

Master of Computer Applications (MCA) | Amrita Vishwa Vidyapeetham, Coimbatore | 2022–2024